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# Exploratory Study on Sustainability in Agile Software Development

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**Abstract.** Sustainability is continuously gaining attention from different software research disciplines and organizations because of the impacts of human activities on the planet aided by software products and services. The process of creating and developing these software products and services through agile has gained significant attention with different guidelines and frameworks proposed to support sustainability. However, there is the challenge of a few concrete illustrations that exemplify how these software sustainability design guidelines and frameworks have been applied in agile practice, specifically Scrum by software development practitioners. This creates no common ground for software development practitioners to understand their role in promoting sustainability during software design and development. This paper explores integrating sustainability into agile software requirement gathering, focusing on the Scrum framework. The outcome presented in this study represents early results from an ongoing case study with an agile development team in the industry.

**Keywords:** Agile, Scrum, requirement, sustainability, software design, SusAF

## 1 Introduction

Sustainable development (SD) and sustainability have gained importance in research with continuously increasing populations and decreasing resources [1]. The continuing aim of SD is to maintain the planet's livability for the upcoming generations [2]. A research study involving selected software developers found that only a few considered sustainability as significant, suggesting a need for more awareness regarding the sustainability impacts of software development [3]. In software engineering, sustainability impacts can be viewed from the following dimensions according to Penzenstadler et al. [4]: Environmental (the use and maintenance of natural resources); Economical (maintaining financial capital); Social (factors that affect the interaction between a group of people such as trust, equality, inclusiveness and diversity); Individual (well-being of individuals and equal access to services); Technical (ensuring that software systems are used for an extended period and can evolve to meet changing requirements).

Research has shown that many software development companies and organizations design their software products and services without considering the sustainability

imapcts [5]. Instead, most companies focus on economic gains by looking at avenues for cutting development and resource costs to reduce time to deliver software products and services. This led to the wide adoption of agile in software development [6], such as Lean Software Development (LSD), Scrum, and Extreme Programming (XP). Agile methods help detect faulty code, embrace changing requirements, and deliver software products and services quickly [7]. From the lens of sustainability in agile software development, Eckstein and O. Mel [8], [9] explored how agile manifesto can be tailored to support social, economic, and environmental dimensions of sustainability. However, there have been less research exemplifying how this is done in companies. Bambazek et al. [10] investigated the elements of the scrum framework to address sustainability. The study outcome [10] shows software development practitioners see good potential for integrating sustainability impacts into the scrum Framework.

Our study builds on these studies to explore avenues for making software development practitioners (agile development team) aware of sustainability and integrating sustainability impacts into the scrum framework. The rest of the work is structured as follows: background study is in section 2. The study design is detailed in section 3. Results with discussion are presented in section 4. Concluding remarks and suggestions for future work are in section 5.

## 2 Background

Agile methodology allows for close collaboration with stakeholders during software project, and new requirements may be accepted even later in the project [15]. Despite the flexibility that agile offers for gathering and implementing software requirements, there are fewer tools for sustainability support to agile development teams. The purpose of requirement engineering (RE) is to implement a "system requirements document" to share knowledge, while agile development relies on achieving a similar goal by communicating with customers face-to-face [13]. RE focuses on modeling, conveying, and documenting software system requirements and the environment in which the system operates [14]. However, most applications of RE steps in agile software development do not cover sustainability because of a lack of sustainability knowledge and understanding. This lack of sustainability awareness by software development practitioners and companies is the reason for the development of sustainability awareness framework (SusAF) [16]. SusAF is a question-based tool coving different topics on the five dimensions of sustainability (Table 1) to help software development practitioners identify the potential sustainability impacts of software products and services.

**Table 1.** Topics covered in each sustainability dimension of SusAF

| Dimensions | Topics |
|---|---|
| Social | Sense of Community, Trust, Inclusiveness and Diversity, Equality, Participation and Communication |
| Individual | Health, Lifelong learning, Privacy, Safety, Self Awareness and Free will |
| Environmental | Material and Resources, Soil, Atmospheric and Water Pollution, Energy, Biodiversity and Land Use, Logistics and Transportation |
| Economic | Value, Customer Relationship Management, Supply chain, Ecosystem, Governance and Processes, Innovation and R&D |
| Technical | Maintainability, Usability, Extensibility and Adaptability, Security, Scalability |

Researchers have shown significant interest in matters relevant to agile software development, which is apparent from many published scientific articles [17]. However, agile practitioners encounter a broad scope of challenges, including sustainability [18]. Sustainability can be seen from two perspectives: sustainable software and sustainability by software. The former concerns the software that is not adversely impacting the economy, society, or the environment by its deployment and production, while the latter concerns utilizing software to accomplish sustainable goals [19]. This study investigates how to incorporate sustainability impacts in agile software development with a starting point of the scrum product backlog. Product backlogs are lists of tasks (unresolved bugs, user stories, chores, etc.) used by software teams for coordinating work, and developers work on the items at the top of the backlog when developing new features, updating existing features, and fixing bugs [20]. We (authors) concentrated on the scrum product backlog as a starting point because that is the source of requirements for any agile software development within the scrum framework.

## 3 Study Design

This research details an ongoing case study for a software project (an online marketplace for service seekers and service providers) with software development practitioners (agile development team) in a company (software development company). The goal of the collaboration with the company is to improve the sustainability awareness among software development practitioners in agile software development processes (scrum). The information about the job title and years of experience in software development of the agile development team involved in the case study are detailed in Table 2.

**Table 2.** Agile Team Background

| No | Job Title | Years of Experience |
|---|---|---|
| 1 | Product Owner | 6 |
| 2 | Programmer | 6 |
| 3 | UI Specialist | 5 |
| 4 | Software Developer/Architect | 6 |
| 5 | IT Manager | 4 |
| 6 | Team Lead (Scrum Master) | 9 |

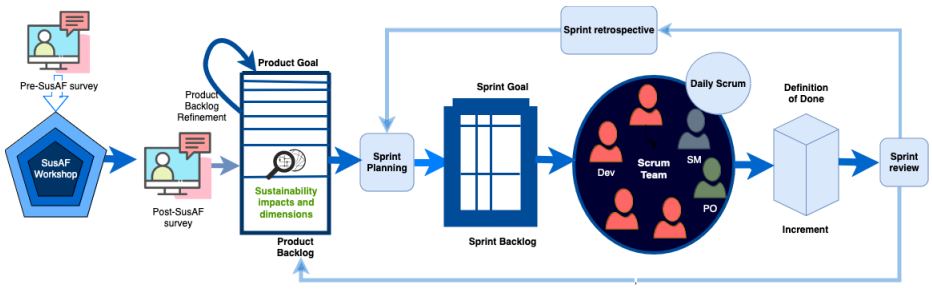


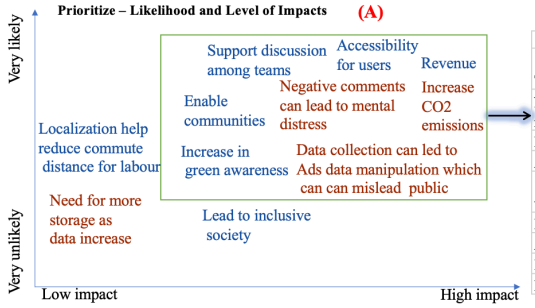

**Figure 1.** Overview of the research process

The study process as shown in Figure 1 begins with a pre-SusAF survey conducted with the agile development team to assess their level of sustainability awareness in software

development. Subsequently, SusAF workshop is conducted to help the agile development team identify the potential impacts of the software project. Next, a post-SusAF survey was conducted to identify the changes in sustainability awareness of the agile development team. Thematic analysis is then performed on the pre and post-SusAF survey data to identify emerging themes. Finally, the researchers and agile development team explored ways of integrating sustainability into the agile scrum framework (product backlog) based on results from the SusAF workshop.

## 4 Results

The agile team utilized SusAF to identify potential sustainability impacts of the software project as shown in Figure 2. Figure 2A (red color text: negative impacts, blue color text: positive impacts) highlights how the agile development team grouped all identified impacts based on the likelihood that the impacts will occur with high or low impact. For example, in Figure 2A SusAF helped the agile development team to identify that potential negative comments from users of the software project could cause other users mental distress (individual dimension), and data gathering within the software project could potentially lead to advertising data manipulation (social dimension).




(B)

| Key Focus Sustainability Impacts | | |
|---|---|---|
| **Top Priority** | **Sustainability Dimension** | **Impact** |
| Negative comments can lead to mental distress | Individual | Negative |
| Reduce CO2 emissions | Environmental | Negative |
| Ad can mislead public | Social | Negative |
| Support discussion among teams | Social | Positive |
| Enable communities | Social | Positive |
| Accessibility to support users | Individual | Positive |
| Increase in green awareness | Social | Positive |
| Revenue | Economic | Positive |

Figure 2. Identified potential sustainability impacts of the software project

Figure 2B shows the results of the agile development team prioritization discussion for the identified sustainability impacts across the sustainability dimensions. Some of the key focus covers measure and reduction of CO2 emissions associated with development of the software project. The agile development also explored accessibility requirements for users, created requirements for a green awareness campaign module to improve user (service providers and service seekers) sustainability awareness and the implementation of content filtering to reduce hate comments on user profile pages.

| User Story ID | Title | Description | Impact Indicator | Sustainability Dimension | Iteration | Owner | Legend |
|---|---|---|---|---|---|---|---|
| 1 | Register web | As WSP/WSS/WC I want to register | Enable communities | Social | 1 | Ben | WSS = Web Service Seeker |
| 2 | Login web | As WSP/WSS/WC I want to login | Increase in green awareness | Environmental | 1 | Ben | WSP = Web Service Provider |
| 3 | Register mobile | As MSP/MSS/MC I want to register | Enable communities | Social | 1 | Ben | WC = Web Company |
| 4 | Login mobile | As MSP/MSS/MC I want to login | Increase in green awareness | Environmental | 1 | Ben | MSP= Mobile Service Provider |
| 5 | Personal service | As WSP I want to add personal services to profile | Increase in green awareness | Environmental | 1 | Ben | MSS= Mobile Service Seeker |
| 6 | Create team | As WSP I can create team with other members | Enable communities | Social | 1 | Ben | MC= Mobile Company |
| 7 | Join team | As WSP I can join other teams | Support discussion among teams | Social | 1 | Ben | |
| 8 | Hired WSP | As WSP I want to get hired by WSS | Revenue | Economic | 2 | Ben | |
| 9 | Bidding | As WSP I can bid for offers by WSS | Revenue | Economic | 2 | Ben | |
| 10 | Hiring WSS/ WC | As WSS/WC I want to hire team or individual | Revenue | Economic | 2 | Ben | |

Figure 3. Sustainability impacts and dimensions linked to each item in the software project product backlog

Subsequently, the agile development team decided to integrate these identified and prioritized sustainability impacts into the scrum product backlog (Figure 3) to ensure that sustainability is considered during the development process in each sprint. For the purpose of anonymity, Ben was used in Figure 3 as the product owner. The researchers examined the product backlog (Figure 3) and discussed them with the agile development team to understand the rationale behind linking each sustainability impact to the product backlog items. The agile development team clarified that all sustainability impacts used in the product backlog were based on the results from SusAF analysis (see Figure 2), which enabled them to focus on high-impact sustainability concerns likely to occur. This process facilitates prioritizing the most critical sustainability impacts during the software development phase.

### 4.1 Pre and Post-SusAF Survey Results

The results of the pre and post-SusAF survey summarized in Table 3 presents the changes in sustainability awareness of the agile development team. The agile development team were asked to list the top two things that come to mind about sustainability in their work during agile (scrum) software development.

Table 3. Pre and post- SusAF survey data from the agile development team

| **Pre-SusAF survey** | | **Post-SusAF survey** | |
|---|---|---|---|
| ***First*** | ***Second*** | ***First*** | ***Second*** |
| Efficiency | Product success | Efficient product to reduce carbon footprint | Inclusiveness and diversity |
| Sustainable pace | Working software | Efficient coding | My health and mood at work |
| Customer satisfaction | Good product | Easy-to-use and maintainable software products | Design for efficiency and reusability |
| Quality | Good performance based on user requirements | Judicious use of resources and project carbon footprint | Support a sense of community in software product |
| Good resource allocation | Great products for our customers = Revenue | Design products that are environmentally efficient in resource consumption | The mental well-being of the final users of our products |
| Usability | Performance | Efficient use of resources among teams | I will now help customers become aware of their CO2 impacts |

The pre-SusAF survey results with the emerged themes indicated the agile development team's focus on the technical sustainability dimension and traditional indicators of success in Agile development. However, the post-SusAF survey results show a shift in their sustainability perception to include more socially conscious and environmentally friendly considerations such as reducing carbon footprint, inclusiveness and diversity, design for efficiency and reusability. Furthermore, after the post-SusAF survey, the agile development team explained to the researchers some of the decisions and actions influenced by their new understanding of sustainability such as:

- Working with the researchers to create a tool for sustainability in agile software development.

- Rethink and evaluate their software development process to incorporate the SusAF workflow.
- Find ways to create a guideline that can facilitate better integration of SusAF for their designers

These results indicate a promising future for better sustainability consideration among the agile team during software design and development.

### 4.2 Discussion

Several sustainability guidelines and frameworks for software requirement elicitation are proposed in different research for software development practitioners to apply in the industry. However, very few concrete examples show how these different sustainability guidelines and frameworks were applied in practice in agile (Scrum) software development of products and services. The early results from the SusAF workshop with the product backlog which includes sustainability impacts and dimensions show there is the potential to consolidate existing research towards transfer of research to practice in agile software development. This can help improve the current state of sustainability practice by software development practitioners.

The collaboration between the researchers and the agile development team during the ongoing case study highlights the issues encountered by software development practitioners when considering sustainability in agile software requirements and development such as a lack of awareness about sustainability and a poor understanding of what sustainability means in software design and development as shown the pre-SusAF survey result (Table 3). This lack of sustainability understanding in software development is also reported in a different study [21]. Another study [22] reported that software development practitioners lack the knowledge and methodological support for acting as facilitators on sustainability issues in their software system.

The SusAF workshop intervention indicated promising results because the sustainability awareness in software development of the agile development team from the post-SusAF survey results showed a transition from a focus on technical sustainability to a holistic view covering the environmental, social and individual sustainability dimensions. Therefore, tool support is required to support and guide software development practitioners for sustainability in agile software development. The results of this study indicate the potential to create sustainability awareness among software development teams in the industry.

### 4.3 Threat to Validity

The results presented in this research are early results from a case study with one company which can limit the generalization of results. However, this study's outcome in terms of the challenges faced by software development practitioners on sustainability integration into software development is also reported in other studies [22] [23], [24].

## 5 Conclusion

Agile methodologies provide the frameworks for software development practitioners in most ICT companies to design and develop software products and services that have

a far-reaching sustainability effect on the users, society, and the environment. Software development practitioners must be empowered to take responsibility for the sustainability impacts of software systems through design decisions in development and deployment. Consequently, it is important to infuse sustainability concerns into agile frameworks such as scrum to promote sustainability awareness among software development practitioners. The results presented in this study are early results from an ongoing case study demonstrating the potential to incorporate sustainability concerns in scrum framework (product backlog). The researchers plan to continuously work with agile development team to create a toolkit covering product backlog and planning, daily scrum, sprint review and retrospective. The aim is to guide software development practitioners in recognizing and improving the sustainability impacts of the different design choices of software systems on sustainability.

## References

[1] B. Penzenstadler and A. Fleischmann, "Teach Sustainability in Software Engineering?," in *24th IEEE-CS Conference on Software Engineering Education and Training (CSEE&T)*, 2011. doi: 10.1109/CSEET.2011.5876124.

[2] C. Manteuffel and S. Ioakeimidis, "A systematic mapping study on sustainable software engineering: A research preview," *9th SC RUG 2011-2012*, p. 35, 2012.

[3] S. Oyedeji, H. Shamshiri, J. Porras, and D. Lammert, "Software Sustainability: Academic Understanding and Industry Perceptions," in *Software Business*, X. Wang, A. Martini, A. Nguyen-Duc, and V. Stray, Eds., in Lecture Notes in Business Information Processing. Cham: Springer International Publishing, 2021, pp. 18–34. doi: 10.1007/978-3-030-91983-2_3.

[4] B. Penzenstadler and H. Femmer, "A generic model for sustainability with process- and product-specific instances," in *Proceedings of the 2013 workshop on Green in/by software engineering*, in GIBSE '13. New York, NY, USA: Association for Computing Machinery, Mar. 2013, pp. 3–8. doi: 10.1145/2451605.2451609.

[5] Oscar Antonio Alvarez Galán, José Luis Cendejas Valdéz, Heberto Ferreira Medina, Gustavo A. Vanegas Contreras, and Jesús Leonardo Soto Sumuano, "Proposal of a Sustainable Agile Model for Software Deevelopment.pdf," *Int. J. Adv. Comput. Sci. Appl.*, 2020.

[6] N. Rashid and S. U. Khan, "Developing green and sustainable software using agile methods in global software development: Risk factors for vendors," *ENASE 2016 - Proc. 11th Int. Conf. Eval. Nov. Softw. Approaches Softw. Eng.*, no. Enase, pp. 247–253, 2016, doi: 10.5220/0005913802470253.

[7] E. and J. G. Wrubel, "Contracting for Agile Software Development in the Department of Defense: An Introduction," *Tech. Rep. CMUSEI-2015- TN-006 Carneige Mellon Univ.*, 2015.

[8] J. Eckstein and C. de O. Melo, "Sustainability: Delivering Agility's Promise," in *Software Sustainability*, C. Calero, M. Á. Moraga, and M. Piattini, Eds., Cham: Springer International Publishing, 2021, pp. 215–241. doi: 10.1007/978-3-030-69970-3_9.

[9] C. de O. Melo, "Another Purpose for Agility: Sustainability," in *Agile Methods*, P. Meirelles, M. A. Nelson, and C. Rocha, Eds., Cham: Springer International Publishing, 2019, pp. 3–7. doi: 10.1007/978-3-030-36701-5_1.

[10] P. Bambazek, I. Groher, and N. Seyff, “Sustainability in Agile Software Development: A Survey Study among Practitioners,” in *2022 International Conference on ICT for Sustainability (ICT4S)*, Jun. 2022, pp. 13–23. doi: 10.1109/ICT4S55073.2022.00013.

[11] M. Endres, P. M. Bican, and T. Wöllner, “Sustainability meets agile: Using Scrum to develop frugal innovations,” *J. Clean. Prod.*, vol. 347, p. 130871, May 2022, doi: 10.1016/j.jclepro.2022.130871.

[12] N. Saher, F. Baharom, and R. Romli, “Identification of Sustainability Characteristics and Sub-Characteristics as Non-Functional Requirement for Requirement Change Management in Agile,” 2020. Accessed: Apr. 05, 2024. [Online]. Available: https://www.semanticscholar.org/paper/Identification-of-Sustainability-Characteristics-as-Saher-Baharom/f6a38e7bdb6349075f1929bdc37b3f71a6194081

[13] A. De Lucia and A. Qusef, “Requirements Engineering in Agile Software Development},” *J. Emerg. Technol. Web Intell.*, 2010.

[14] F. Paetsch and F. Maurer, “Requirements Engineering and Agile Software Development,” pp. 1–6, 2003.

[15] M. Fowler and J. Highsmith, “The agile manifesto,” *Softw. Dev.*, vol. 9, no. August, pp. 28–35, 2001, doi: 10.1177/004057368303900411.

[16] “SusAF - the sustainability awareness framework,” *Karlskrona Manif. Group*, 2019.

[17] T. Dingsøyr, S. Nerur, V. Balijepally, and N. B. Moe, “A decade of agile methodologies: Towards explaining agile software development,” *J. Syst. Softw.*, vol. 85, no. 6, pp. 1213–1221, 2012, doi: 10.1016/j.jss.2012.02.033.

[18] P. Gregory, L. Barroca, H. Sharp, A. Deshpande, and K. Taylor, “The challenges that challenge: {Engaging} with agile practitioners’ concerns,” *Inf. Softw. Technol.*, vol. 77, pp. 92–104, 2016, doi: 10.1016/j.infsof.2016.04.006.

[19] N. Condori-Fernandez and P. Lago, “Towards a Software Sustainability-Quality Model: Insights from a Multi-Case Study,” in *2019 13th International Conference on Research Challenges in Information Science (RCIS)*, May 2019, pp. 1–11. doi: 10.1109/RCIS.2019.8877084.

[20] “The Product Backlog | IEEE Conference Publication | IEEE Xplore.” Accessed: Feb. 11, 2024. [Online]. Available: https://ieeexplore-ieee-org.ezproxy.cc.lut.fi/abstract/document/8812076

[21] S. Oyedeji, A. Seffah, and B. Penzenstadler, “A catalogue supporting software sustainability design,” *Sustainability*, vol. 10, no. 7, pp. 1–30, 2018, doi: 10.3390/su10072296.

[22] L. Duboc *et al.*, “Requirements engineering for sustainability: an awareness framework for designing software systems for a better tomorrow,” *Requir. Eng.*, vol. 25, no. 4, pp. 469–492, Dec. 2020, doi: 10.1007/s00766-020-00336-y.

[23] R. Heldal et al., “Sustainability competencies and skills in software engineering: An Industry perspective,” Journal of Systems and Software, p. 111978, 2024

[24] S. Betz et al., “Lessons Learned from Developing a Sustainability Awareness Framework for Software Engineering using Design Science,” ACM Transactions on Software Engineering and Methodology, 2024.